

James Horrigan

Full Stack AI Engineer | Python, AWS, React, Next.js | AI Product Builder

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[Portfolio](#) | [LinkedIn](#) | [Github](#)

PROFESSIONAL SUMMARY

Full stack engineer with 30 years in technology, across many industries and company sizes, as both a contractor and permanent employee. Current work is AI-native. Most recently built and launched a UK trade price-comparison platform end to end, delivering sub-second search and browse across a large catalogue, cutting its worst endpoint from a 21-second average to under **0.4** seconds, and raising automatic product classification from 16% to **69%** using vector embeddings and retrieval.

Earlier delivery includes a **70%** reduction in customer call volumes for a debt management company, **£1.3** million in hiring cost savings for a transport operator, and the ingestion of **300** million products for a product search platform. Founder of XPLORATECH.AI, contracting since 2023, building production software with agentic AI development tooling alongside the engineering standards and verification that keep that output reliable.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Frameworks & Libraries: React, Next.js, Django, Flask, Tailwind CSS, MUI, Pydantic

AI and Machine Learning: Large Language Models (LLMs), Generative AI, vector embeddings, vector search, pgvector, semantic search, Retrieval-Augmented Generation (RAG), prompt engineering, OpenAI API, AWS Bedrock, Natural Language Processing (NLP), spaCy, AI-assisted and agentic development, Claude Code

Databases: PostgreSQL, Neon, DynamoDB, MySQL, Amazon RDS, database migrations

Cloud and Infrastructure: AWS (Lambda, API Gateway, ECS, Fargate, S3, CloudFront, Cognito, SQS, CloudWatch, VPC, IAM), Docker, Terraform, SST, Infrastructure as Code, serverless architecture, Azure DevOps

APIs and Integrations: RESTful API development, Stripe, OAuth2, n8n, webhooks, OpenAPI, JWT

Testing and Delivery: pytest, Vitest, Playwright, CI/CD, Git, Agile delivery, technical leadership, solution architecture

WORK EXPERIENCE

Founder and Full Stack AI Engineer | XPLORATECH.AI | Jul 2023 - Present | Remote, UK

- Provide full-stack engineering and applied AI services to agencies and product teams, contracting across the debt management and construction supply sectors.
- Deliver production systems end to end across front end, back end and infrastructure, specialising in vector search, retrieval pipelines and language-model features in production.
- Apply agentic AI development tooling throughout delivery, with written standards, workstream isolation and automated verification to keep output reliable.

Founding Engineer | MYTRADE TECHNOLOGIES via XPLORATECH.AI | Mar 2026 - Aug 2026 | Remote, UK

- Built and launched MYTRADEBARGAINS, a UK trade price-comparison platform from scratch, using a Python serverless API on AWS, five Next.js applications, PostgreSQL and DynamoDB, all infrastructure as code, live and earning affiliate revenue within weeks of the first commit.
- Cut the worst-performing endpoint from a 21-second average to under 0.4 seconds, removing 73% of all slow requests; reduced category browsing from 5.5s to 0.9s and settled search at a 537ms average.
- Rebuilt product categorisation on vector embeddings and retrieval, raising confident automatic classification from 16% to 69% and cutting uncertain outcomes from 29% to 3%.

- Re-architected ingestion onto per-supplier AWS Fargate Spot tasks, cutting compute cost by approximately 70% with resumable, interruption-safe writes.
- Applied LLMs and vector embeddings across supplier discovery, record enrichment, catalogue classification and semantic search, keeping model spend negligible across the catalogue's lifetime.
- Provisioned and ran the back-office automation platform on Terraform-managed AWS EC2 running n8n with no SSH access, carrying around 15 version-controlled workflows for cost reporting, monitoring, lead generation and marketing.

Full Stack Software Engineer | Intrum via XPLORETECH.AI | Oct 2023 - Jun 2025 | Remote, UK

- Led full-stack development of a new customer self-service portal, reducing contact centre call volumes by 70% after launch.
- Architected customer-facing debt management portals using React, Python, API Gateway and AWS serverless architecture, serving enterprise-scale client bases.
- Implemented OAuth2 authentication and white-labelling across multiple regulated client deployments, supporting client-specific branding and authentication flows.
- Delivered content delivery infrastructure on AWS Lambda and CloudFront, reducing page load latency by over 40%.
- Built AI-assisted agent tooling presenting call transcripts and sentiment analysis to contact centre agents, improving operational throughput.

CTO and Software Engineer | CiiVSOFT | Jan 2019 - Jul 2023 | Remote, Warrington, UK

- Led the engineering function as CTO, designing a scalable Python architecture for concurrent data ingestion from enterprise applicant tracking systems, using Django, Celery, Docker and AWS with asynchronous and multiprocessing techniques to overcome IO and CPU-bound limits.
- Built an in-house NLP resume parser using Flask and spaCy, removing a costly and unreliable third-party dependency, processing thousands of job applications per hour for the first enterprise client.
- Delivered £1.3 million in hiring cost savings for a transport operator, and led onboarding for major European clients including Alstom, Agoda, Babcock, EDP and Glovo.
- Replaced key infrastructure with serverless microservices on EventBridge, Step Functions, Lambda and DynamoDB, and built the engineering team, running code reviews, mentoring and Agile delivery.

Data Platform Engineer | Priceseacher.com | Jan 2017 - Dec 2018 | London, UK

- Engineered high-throughput Python pipelines ingesting 300 million live products and price changes daily, direct from merchants, with a feed processor handling any format (CSV, XML, JSON, text) from URL and FTP sources using parallel processing and queues.
- Built bespoke feed processors for Amazon, eBay and Billiger, and rebuilt the downloader to handle frequent downloads from around 1,000 clients.
- Led two large data migrations: 300 million products from Elasticsearch to DynamoDB with processors rewritten from PHP to Python, and a field value change applied across a database of 1.3 billion products.

Earlier career | Capita Customer Management, Jan 2008 - Dec 2016 (Web Administrator / Communications Manager, then Innovations Technician) - built real-time operational tooling, reporting dashboards and internal prototypes across multiple business sites. Marconi PLC, Aug 1996 - Dec 2006 (Software Engineer) - embedded C software for payphone systems, plus automated testing workflows and engineering utilities.

EDUCATION AND CERTIFICATIONS

AWS Certified Developer - Associate, Amazon Web Services | Oct 2023 | Credential RM8RZ6510NQ1QBS4

HND in Software Engineering, St. Helens College | 1997 - 1999

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